AquaBlend

Deakin University — Capstone Project

In Search of Cluster Analysis

A survey and evaluation of clustering techniques for AquaBlend data

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v0.1	August 27, 2026	B. M. Tran	Initial document structure.

Abstract

One paragraph covering: the problem this document addresses, the clustering techniques surveyed, how they were evaluated, and the headline finding or recommendation for AquaBlend. **[TODO: write once the first technique is complete]**

Contents

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Notation and acronyms

Acronyms

Acronym	Expansion
AI	Artificial Intelligence
UML	Unsupervised Machine Learning
HCA	Hierarchical Cluster Analysis
DHC	Divisive Hierarchical Clustering
AHC	Agglomerative Hierarchical Clustering
NC	Number of Cluster
DR	Dimensionality Reduction
EDA	Exploratory Data Analysis
SOM	Self–Organising Map
GNN	Graph Neural Network
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
SSE	Sum of Squared Error
SC	Silhouette Coefficient

Mathematical notation

Symbol	Meaning
$\mathbf{X}$	Matrix representation of the dataset, rows for samples and columns for features
m	Number of observations
n	Number of features
$\mathcal{C}$	Set of clusters
$\mathcal{C}_i$	i^{th}–cluster of $\mathcal{C}$
$\lvert\mathcal{C}\rvert$	Number of clusters
$d(\cdot,\cdot)$	Distance / dissimilarity measure
$\boldsymbol{\mu}_i$	Centroid (prototype) of cluster $\mathcal{C}_i$
$s(\mathbf{x})$	Silhouette of observation $\mathbf{x}$
$\mathcal{O}(\cdot)$	Big–O asymptotic notation

Conventions

State document-wide conventions here, e.g. vectors are column vectors and set in bold, indices i range over observations and j over clusters, all distances are Euclidean unless stated otherwise, software versions and random seeds used for every run.

1 Introduction

1.1 Motivation

Arising from the desire of improving the project proposal, allowing the core mathematical models to become more proactive, rather than being reactive by nature. In addition, realising the potential of the AquaBlend dataset can be further explored and studied to improve the performance of future Mixed–Integer Nonlinear Programming (MINLP) model, along with insightful patterns undiscovered in the dataset, which is impactful in future research on related problems about water systems based in Melbourne, and possibly adopted to study water systems in other locations.

This document is made to accomodate the needs of summarizing the deliverables of the codebase provided in the project's repository and solid theoretical and practical foundations for exploring and analysing AquaBlend dataset, to facilitate a broad range of functionalities of the project deliverables, including but not limited to, Time–series analysis, Demand Forecasting, Anomaly Detection and Scenarios Generation.

1.2 Objectives

This section outlines questions addressed in this document.

1. What potential clustering methods and dimensionality reduction techniques can be applied to specifically study the AquaBlend dataset?
2. How well the selected methods perform on the AquaBlend dataset subject to certain standard evaluation metrics/scores?
3. Can there be an appropriate Exploratory Data Analysis and Data Mining pipeline to discover patterns in the AquaBlend dataset? If so, can it be standardised to apply to related problems? Otherwise, why is it impossible?
4. Is "Manifold Hypothesis" valid in the dataset? If so, what can be done to improve the clustering results? Otherwise, is there strong statistical evidence attained from the AquaBlend dataset?

1.3 Scope and limitations

Cluster Analysis is a broad research area, thereby we limit ourselves to discussing on specific methods and problems associated, mentioning other methods without going in details. However, this document will be extended to include more clustering methods in the upcoming work.

For the current version, we limit ourselves to those methods and techniques particularly useful to uncover knowledge from AquaBlend dataset. In recent future, we shall improve the pipeline to become adaptable to discover knowledge from relevant datasets aiming to optimise Operational Water Resources Management sector.

More importantly, this document is not a comprehensive survey of all existing studies of Cluster Analysis, and thereby, there may exist more effective and appropriate techniques yet to mentioned within this document. Hence, this document can only be used as a reference for a collection of potential methods, and how they can be practically implemented to solve certain problems.

1.4 Document structure

The document is organised as follows.

1. Survey recent development in Cluster Analysis, see **Sect. 2**.
2. Introduce dataset used (see **Sect. 3**) and experimental setup to benchmark different clustering strategies (see **Sect. 4** and **App. A**).
3. Deliver the Exploratory Data Analysis process and Data Mining pipeline conducted, see **Sect. 5**.
4. Discuss on various problems associated with data dimensionality and possible solutions, see **Sect. 6**.
5. Summarise the theory behind the implementation of clustering methods, see **Sect. 7**.

2 Background

2.1 Cluster analysis in brief

Clustering analysis refers to the process of exploring the dataset to uncover the latent structure and underlying relationships among data points, commonly known as unsupervised machine learning (UML) in the AI community, and essentially important in data mining and knowledge discovery [20]. It comprises of all clustering methods that are capable of handling (usually) high–dimensional data and grouping them by the utilisation of several (usually) quantitative measurements metrics, see [10] for a brief history of development. In this document, we shall refer to a clustering method as [24]:

- Minimising within–cluster (intra–cluster) distance/dissimilarity.
- Maximising inter–cluster distance/dissimilarity.
- Measurement of distance/dissimilarity must be well–defined and practically meaningful.

Formally, let us claim the definitions of a valid mathematical metric and a clustering method.

Definition 2.1 (A mathematical metric [19])**.** Given two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, a (mathematical) metric is given by $d : \mathbb{R}^n \times \mathbb{R}^n \mapsto \mathbb{R}$, which satisfies:

- Identity and Positivity: $d(\mathbf{x}, \mathbf{y}) > 0$ if $\mathbf{x} \neq \mathbf{y}$, and $d(\mathbf{x}, \mathbf{y}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{y}$
- Symmetry: $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$
- Triangle inequality: $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y}), \forall \mathbf{z} \in \mathbb{R}^n$

Definition 2.2 (Clustering method [20])**.** Given a dataset $\mathbf{X} \triangleq \{\mathbf{x}_i \in \mathbb{R}^n\}_{i=1}^m$. A clustering method is defined as $\phi_d : \mathbf{X} \mapsto \mathcal{C}$, such that:

$$\bigcap_{i=1}^{|\mathcal{C}|} \mathcal{C}_i = \emptyset \quad \text{and} \quad \bigcup_{i=1}^{|\mathcal{C}|} \mathcal{C}_i = \mathbf{X} \tag{1}$$

and ϕ_d is (usually) aiming to:

$$\min \sum_{i=1}^{|\mathcal{C}|} \sum_{\substack{\mathbf{x}_j, \mathbf{x}_k \in \mathcal{C}_i \\ \mathbf{x}_j \neq \mathbf{x}_k}} d(\mathbf{x}_j, \mathbf{x}_k) \quad \text{and} \quad \max \min_{1 \leq i < j \leq |\mathcal{C}|} \left\{ \min_{\substack{\mathbf{x}_i \in \mathcal{C}_i \\ \mathbf{x}_j \in \mathcal{C}_j}} d(\mathbf{x}_i, \mathbf{x}_j) \right\} \tag{2}$$

where $d(\cdot, \cdot)$ is a *dissimilarity measure*, need not to be a metric as in **Def. 2.1**.

Although different literatures proposing different claims of standard process of cluster analysis [10], though sharing a lot in common, we shall adopt the process written in [24].

1. Feature extraction and selection: opt out representative features of the dataset.
2. Clustering algorithm design: select appropriate methods for the data characteristics and problems' constraints.
3. Cluster evaluation/validation: quantify the performance of clustering results by use of validity measures.

4. Result interpretation: explain what drives the method to identify the patterns.

Looking from the above definition and formulation of clustering methods, several issues have been identified, which are (mostly) subject to dataset and nature of clustering methods [20]. Regarding dataset, the Curse of Dimensionality leads to indistinguishable distances between points in high–dimensional space [16], which degrades the capability of dissimilarity measure used in the methods. Other hurdles are, including but not limited. to, size of dataset, noise/outliers, and mixed data type. Regarding methods, many of them requires hyperparameter tuning, which is subjective and data–dependent, also but not limited to, computation complexity, convergence speed, and local optima convergence. Therefore, the upcoming clustering techniques have been developed to mitigate some of issues associated with the problems at some extent.

2.2 Families of clustering techniques

This section serves as a brief list of methods' names, for details what each family include, refer to **Sect. 7**. There is no universal agreement on how clustering methods are classified, several articles, e.g. [6, 10, 23] and books, e.g. [9, 21] rely on different criteria to divide these clustering methods into groups depending on strategies implemented. In this document, we adopt the philosophy in [20] to classify the methods into two major groups.

Hierarchical clustering Hierarchical Cluster Analysis (HCA) offers multi–level partitions of the dataset, usually represented using a dendogram, and one can select a threshold to obtain different clustering outcomes. It is further categorised into two groups, i.e. a top–down strategy, known as Divisive Hierarchical Clustering (DHC), and a bottom–up strategy, known as Agglomerative Hierarchical Clustering (AHC). Although differing in construction approaches, they share same components, including dissimilarity measure and linkage criterion. However, DHC is more favorable in practices due to its efficiency.

Partition–based clustering Partitional clustering or iterative relocation algorithms focus on diving the dataset into (nearly–)homogenous groups without the needs of hierarchical structure, aiming to optimise certain criteria functions, can be understood as those given in **Def. 2.2**. It is further categorised into two groups, i.e. hard/crisp clustering and soft/fuzzy clustering. In lieu of considering crisp and fuzzy clustering as two primary groups within partitional methods as in [6, 20], we shall classify crisp clustering methods in more thorough manners and keep fuzzy clustering method as is. Hence, the family of partition–based methods, including but not limited to, graph–theoretic, subspace, density–based, model–based, search–based, SSE–optimised, and miscellaneous clustering methods.

Hybrid clustering this family of method comprises those techniques that are mixture of the other techniques, irrespective of what family they reside in.

In reality, clustering can have any arbitrary shape and distribution, this fact underpins distinct classification of the clustering methods. In brief, there are certain types of clusters that have been extensively studied, e.g. well–separated, prototype–based, graph–based, density–based, and shared–property. For detailed visualisations of these clusters in 2D-space, see [21].

2.3 Related work

Cluster analysis is a fundamental phase in knowledge discovery process, and thus, can be applied to a broad range of real–world applications, e.g. bioinformatics [23], health research [5], and network–related problems [7]. Nonetheless, we limit ourselves to revisit recent work tackling issues and demands associated with water systems in the past few years.

Two applications of clustering methods, i.e. K–Means and Gaussian mixture clustering, were leveraged by [4], which assisted them to detect recurring operational states, or “fingerprints”, using data obtained from wasterwater treatment plants based in Ontario, Canada. Thereafter, successfully identified the associations between these fingerprints and critical events, which allows engineers to identify early signals of critical events from operational conditions. A more closely related research was conducted in Finland, which attempted to model changes in water quality through different stages in a treatment process [11]. They first projected the features onto a lower–dimensional space by the utilisation of Self–organising Map (SOM), and subsequently applied K–Means to cluster the projected data, and successfully identified different prototypes in the data, which implies the variations of water quality through distinct phases conducted in a treatment plant. Indeed, the application of SOM have been widely adopted in water–related research, with two surveys devoted to it in recent years. In particular, [1] systematically reviewed how SOM had been applied to water quality assessment in reservoirs and lakes. Meanwhile, [14] highlighted the improved interpretability of SOM when being merged with a clustering algorithm, as studied in [11], and superiority over other relevant methods in modelling the nonlinear relationship between variables, and can be well visualised in 2D or 3D space.

Aside from the scope of water quality, the majority of the literature focusing on discovery of water demands. More specifically, A comparative study on daily consumption patterns evaluates K–means, agglomerative hierarchical and spectral clustering under the Silhouette Coefficient and the Calinski–Harabasz index [8]. Closer to density–based methods, a modified DBSCAN with automatic optimisation of the neighbourhood radius and explicit detection of noise points has been applied to segment daily consumption profiles [12]. Time–series formulations are also represented, by clustering of consumption series combined with entropy analysis [22] and by clustering of hydrological telemetry series [17]. More recently, a graph–based clustering method, specifically a Graph Neural Network (GNN), was proposed in [3], to leverage the interdependencies between the dataset and external information unavailable to it, which is likely to govern the urban water demands.

A relatively sparse research direction is the problem–driven scenario clustering. Typically, scenarios are generated by sampling from the distributions of the operational dataset, and sometimes overlook the mathematical models, as highlighted in [13]. A recent literature, see [2], not only worked on this problem–driven scenario clustering method, but also emphasised the pivotal role of feature engineering to yield better clustering results. Both of them inspired a more appropriate scenario generation approaches to comply with the real–world cases while considering the mathematical model used. This is essentially important, as the mathematical model will soon be developed to become a stochastic programming model, and in reality, the parameters given to the model are not fixed.

Up to our knowledge, no recent work connects regimes discovered in an operational water–quality feature space to source–blending or source–selection decisions, and we suppose this is a potential and abundant area to discover and study.

3 Data

3.1 Source and provenance

Where the data comes from, who owns it, which extract or version is used, and the date it was pulled.

3.2 Features

Table of features: name, type, unit, range, and whether the feature is used for clustering or held out for interpretation.

3.3 Preprocessing

Cleaning, missing-value handling, encoding, scaling, and dimensionality reduction — everything applied before clustering, in the order applied, so the pipeline is reproducible.

3.4 Exploratory summary

Distributions, correlations, and anything that constrains the choice of technique (cluster shape, density, outliers, scale).

4 Evaluation protocol

4.1 Experimental setup

Hardware, software versions, random seeds, and the number of restarts per run. Identical for every technique.

4.2 Validation metrics

The internal and external validity measures reported for every technique, with the definition of each given once here.

Table 1: Validation metrics reported for every technique.

Metric	Range	Interpretation
Metric	*[a, b]*	*What a high or low value means*

4.3 Choosing the number of clusters

The procedure applied consistently wherever k must be fixed in advance.

4.4 Interpreting and naming clusters

How a resulting partition is turned into something the business can act on, and who signs off on the naming.

4.5 Threats to validity

Known weaknesses of this protocol and how they are mitigated.

5 Exploratory Data Analysis and Data Mining Pipeline

5.1 Exploratory Data Analysis (EDA) methods

5.2 Data Mining pipeline

6 Discussion on Dimensionality problems

6.1 Curse of Dimensionality

6.2 Manifold hypothesis and Manifold learning

6.3 Linear Dimensionality Reduction techniques

6.3.1 <NAME>

Overview *What the technique does, in one paragraph: what it maps from, what it maps to, and what it tries to preserve in doing so. State whether it is linear or nonlinear, and for a nonlinear technique, its relation to the manifold hypothesis of Section 6.*

Assumptions and applicability *What the technique assumes about the data — linearity, local versus global structure, density, scale, noise — and the conditions under which it is a sensible choice for the data described in Section 3. State explicitly whether it is intended for visualisation, for reducing input to a clustering method, or for both: these are different jobs and a technique good at one may distort what the other needs.*

Formulation *The objective being optimised and the mapping obtained, using the symbols declared in the notation table. Cite the original source.*

Out-of-sample and inverse mapping *Two questions, answered explicitly, because the rest of the pipeline depends on both.*

First: is the technique inductive or transductive? That is, does it learn a mapping that applies to observations unseen during fitting, or does it embed only the sample it was given? A transductive technique cannot precede a clustering step in a pipeline that will later see new data.

Second: does an inverse mapping exist? If so, a component can be read back in terms of the measured features; if not, say so, and say where the interpretation of the reduced space will come from instead.

Hyperparameters and tuning *Each hyperparameter, the range searched, and the value selected — with the reason. For neighbourhood-based techniques, state what the neighbourhood size does to the balance between local and global structure, since it is the parameter most likely to change the conclusions drawn from a figure.*

Choosing the number of components *How the target dimension was chosen, and on what evidence: an explained-variance criterion, an intrinsic dimension estimate (see Section 6), a reconstruction error, or the requirements of a downstream method. Choosing two because two is what plots is a legitimate choice for a figure and not for an analysis — if that is the reason, say so.*

Complexity *Time and space complexity, and what that means at AquaBlend data volumes. Note any step that is quadratic in the number of observations, since that is what decides whether the technique is usable on the full dataset or only on a sample.*

Application to AquaBlend data *Implementation: library and version, random seeds, and how the run is reproduced. Then any deviation from the shared preprocessing pipeline of Section 3.3, and why it was necessary — scaling in particular, since most of these techniques are not scale-invariant.*

Results *Two things, in this order.*

Structure preservation: how faithfully the embedding represents the input, reported quantitatively — explained variance, trustworthiness, stress, or reconstruction error, whichever the technique admits — and not by appeal to the appearance of a scatter plot.

Effect on downstream clustering: the validity measures of Section 4.2 for the same clustering method fitted with and without this reduction. This is the part that decides whether the technique earns its place in the pipeline.

Limitations *Where this technique distorted the data, or would be expected to. Distinguish what was observed on our data from what is known from the literature. For nonlinear techniques, state plainly which properties of the embedding must not be read as properties of the data — typically inter-cluster distances, relative cluster sizes, and apparent density.*

Summary *Two or three sentences: the verdict on this technique, and whether it is recommended for visualisation, for the clustering pipeline, for both, or for neither. Phrased so it can be carried directly into Section 8.*

6.4 Nonlinear Dimensionality Reduction techniques

7 Clustering methods

Refer to **Sect. 2** for details of cluster analysis, families of methods, and important properties.

Before going into each family of clustering technique, we shall review two primary concepts, i.e. dissimilarity/similarity measures and validity indices. In short, **dissimilarity/similarity measure** determines of extent of association between data instances, and hence, widely used as the objectives to optimise and assignment of data to clusters [20]. Meanwhile, **validity indices** are typically used to qualify the clustering outcomes, whether our chosen methods produce proper clusters in the given dataset [20].

7.1 Dissimilarity measures

7.2 Validity indices

7.2.1 Silhouette Coefficient

Overview The Silhouette Coefficient (SC) is an internal validity index. It scores a partition from the data and labels alone, contrasting for each observation the closeness to its own cluster with that to the nearest other cluster [9, 21]. It applies to any method returning a crisp partition, making it the common ground of **Sect. 8**, and the water–sector literature already uses it [8].

Definition For $\mathbf{x} \in \mathcal{C}_i$, let $a(\mathbf{x})$ and $b(\mathbf{x})$ be the mean dissimilarity to the rest of its own cluster and to the nearest other cluster,

$$a(\mathbf{x}) = \tfrac{1}{|\mathcal{C}_i|-1} \textstyle\sum_{\mathbf{y}\in\mathcal{C}_i\setminus\{\mathbf{x}\}} d(\mathbf{x},\mathbf{y}), \quad b(\mathbf{x}) = \min_{j\neq i} \tfrac{1}{|\mathcal{C}_j|} \textstyle\sum_{\mathbf{y}\in\mathcal{C}_j} d(\mathbf{x},\mathbf{y}), \quad s(\mathbf{x}) = \tfrac{b(\mathbf{x})-a(\mathbf{x})}{\max\{a(\mathbf{x}),b(\mathbf{x})\}} \tag{3}$$

with $s(\mathbf{x}) = 0$ for a singleton $\mathcal{C}_i$. The SC of a partition is the mean of $s(\mathbf{x})$ over all m observations [9, 18, 21].

Properties $s(\mathbf{x}) \in [-1, 1]$ and higher is better, near 1 the observation lies far closer to its own cluster than to any other, near 0 between two clusters, negative when it is arguably misassigned. It is not corrected for chance, is defined for $2 \leq |\mathcal{C}| \leq m-1$, and needs only $d(\cdot,\cdot)$, so it holds for a dissimilarity that is no metric in the sense of **Def. 2.1**.

Applicability It accepts any data type for which $d(\cdot,\cdot)$ is defined and serves every family of **Sect. 2.2** returning a crisp partition, a soft partition must be defuzzified first. Observations labelled noise have no $a(\mathbf{x})$: excluding them scores only the points a method was confident about.

Computation and complexity Obtaining a and b for every observation requires every pairwise dissimilarity, i.e. $\mathcal{O}(m^2 n)$ time and $\mathcal{O}(m^2)$ memory where the matrix is held.

Application to AquaBlend data *Which methods are scored under it, whether it also drives the choice of $|\mathcal{C}|$ in Section 4.3, and whether it is computed on the full dataset or a fixed sample.*

Behaviour Since a enters negatively and b positively, the index rewards clusters compact and mutually distant under $d(\cdot,\cdot)$, which is the geometry a centroid–based method optimises. It therefore does not rank families neutrally, favouring the convex, comparably sized clusters of **Sect. 7.4.1** over elongated or variable–density structure [9, 21].

Limitations Beyond that bias above, it is uninformative for degenerate partitions. Misleading where cluster sizes are unbalanced, the mean being dominated by the largest cluster. Suffering from Curse of Dimensionality, where the concentration of distances of **Sect. 6** compresses a and b together and $s(\mathbf{x})$ towards zero [16].

Summary The SC quantifies cohesiveness and separatability of clusters under $d(\cdot,\cdot)$.

7.3 Hierarchical clustering methods

7.3.1 <NAME>

Overview *What the algorithm does, in one paragraph, and the family it belongs to (see Section 2.2).*

Assumptions and applicability *What the algorithm assumes about cluster shape, density, scale and noise; the conditions under which it is a sensible choice for the data described in Section 3.*

Formulation *Objective function and update rule, using the symbols declared in the notation table. Cite the original source.*

Hyperparameters and tuning *Each hyperparameter, the range searched, and the value selected — with the reason.*

Complexity *Time and space complexity, and what that means at AquaBlend data volumes.*

Application to AquaBlend data *Implementation, any deviations from the shared preprocessing pipeline, and why they were necessary.*

Results *Metrics from Section 4.2, cluster profiles, and supporting figures.*

Limitations *Where this method failed or would be expected to fail on this data.*

Summary *Two or three sentences: the verdict on this method, carried into the comparison section.*

7.4 Partition–based clustering methods

7.4.1 K–Means

Overview K–Means is the canonical crisp partitional method, belonging to the SSE–optimised family of **Sect. 2.2**. It represents every cluster by a centroid and assigns each observation to the nearest one, so a partition is described by $|\mathcal{C}|$ prototypes rather than by the observations [21]. It serves as the baseline the other methods are compared against [20, 24], and has precedent on operational water data [4, 8].

Assumptions and applicability Assignment to the nearest centroid induces a Voronoi partition of the feature space, hence clusters are assumed convex, roughly spherical, and of comparable size and density [9, 21]. It further assumes $|\mathcal{C}|$ known in advance, no noise label, and features on comparable scales, the objective below not being scale–invariant, where these fail, the partition remains well defined but ceases to be meaningful.

Formulation For fixed $|\mathcal{C}|$, K–Means minimises the within–cluster sum of squared error

$$\min_{\mathcal{C}} \sum_{i=1}^{|\mathcal{C}|} \sum_{\mathbf{x}\in\mathcal{C}_i} d(\mathbf{x}, \boldsymbol{\mu}_i)^2, \qquad \text{where} \qquad \boldsymbol{\mu}_i = \frac{1}{|\mathcal{C}_i|} \sum_{\mathbf{x}\in\mathcal{C}_i} \mathbf{x}, \tag{4}$$

the criterion–function reading of **Def. 2.2**, with a centroid standing in for pairwise comparison [21]. The standard algorithm alternates assignment of each $\mathbf{x}$ to the nearest centroid with recomputation of every $\boldsymbol{\mu}_i$ as the mean of its members [9, 24], as neither step increases the objective, it terminates at a local minimum only [6, 15].

Hyperparameters and tuning Three: $|\mathcal{C}|$, fixed by the shared procedure of **Sect. 4.3** rather than tuned per method; the initialisation and the number of restarts, which exist only because the objective is non–convex, so the best objective over several restarts is retained [6, 24].

Complexity One iteration costs $O(m|\mathcal{C}|n)$ time and $O(|\mathcal{C}|n)$ space beyond the data [21, 24]: the cheapest method surveyed, and the only one needing no pairwise dissimilarity matrix, so it remains usable at full data volume.

Application to AquaBlend data *Library and version, seed, restarts, and any deviation from the pipeline of Section 3.3, scaling in particular.*

Results *Every index of Section 4.2, the evidence behind the chosen $|\mathcal{C}|$, and the cluster profiles.*

Limitations Local convergence, so the partition depends on initialisation. The centroid is a mean, hence sensitive to outliers, which cannot be labelled noise. Spherical, equal–size bias splits elongated structure and merges structure of uneven density [6] [21]. The Silhouette Coefficient of **Sect. 7.2.1** rewards precisely the geometry K–Means produces, so that pairing flatters it **(Sect. 4.5)**.

Summary The cheap, scale–sensitive baseline, linear in m, adequate wherever regimes are compact and comparable in size. Whether it earns more than that here is carried into **Sect. 8**.

7.5 Hybrid clustering methods

8 Comparative discussion

8.1 Side-by-side results

Table 2: Validation metrics across all techniques surveyed.

Technique	Metric 1	Metric 2	Runtime
Technique one	–	–	–

8.2 Qualitative comparison

Trade-offs the metrics do not capture: interpretability, stability across seeds, sensitivity to preprocessing, operational cost.

8.3 Recommendation

Which technique AquaBlend should adopt, under what conditions, and what would change the recommendation.

9 Conclusion and future work

9.1 Conclusion

The answer to each objective listed in Section 1.2, in the same order.

9.2 Future work

Techniques not yet covered, open questions, and what the next contributor to this document should pick up first.

A Reproducibility

A.1 Environment

Language, library and package versions, and how to recreate the environment.

A.2 Running the analysis

Commands to reproduce every result in this document, and where the code and outputs live in the repository.

B Supplementary results

Full metric tables, additional figures, and parameter sweeps that would interrupt the main narrative.

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Note here any use of generative AI tools, stating what they were used for and how the output was verified.

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